

Advanced Mathematics Applications

London Wallace

Semester 2, 2026

Contents

Preface	v
0.1 More detailed overview of the subject.	v
0.2 Table of Symbols	v
 I Essentials of Multivariable Calculus	 1
1 Vectors, Vector Valued Functions, and Functions of Several Variables	3
1.1 The Vector Space $\mathbb{R}^n$	3
1.2 Vector Valued Functions	5
1.3 Functions of Several Variables	5
 2 Limits and Continuity	 7
2.1 Functions of One Variable	7
2.1.1 Divergence to, and Limits at, Infinity	10
2.1.2 One Sided Limits	10
2.2 Vector Valued Functions	11
2.3 Techniques for Evaluating Limits	11
2.4 Limits of Multivariable Functions	12
2.5 Continuity	13
 3 Derivatives	 15
 4 Applications of the Derivative	 17
 5 The Riemann Integral	 19
 6 Double and Triple Integrals	 21
 7 Path and Surface Integrals	 23
 8 Vector Fields	 25

Preface

THESE NOTES follow the course *Advanced Mathematics Applications*, taught by Lyle Noakes at the University of Western Australia in Semester 2 of 2026. The unit teaches advanced techniques of calculus, including multivariable calculus, Green's, Stoke's and divergence theorem, complex analysis, ordinary and partial differential equations, and Sturm-Liouville theory.

0.1 More detailed overview of the subject.

THESE NOTES are about something but with more detail.

0.2 Table of Symbols

Symbol	Meaning
$\forall$	For every <i>or</i> for all
$\exists$	There exists
$\in$	In <i>or</i> is an element of
$\subset$	Is a proper subset of
$\subseteq$	Is a subset of
$\cup$	Union with
$\cap$	Intersection of
$\setminus$	Not <i>or</i> without
$\emptyset$	Empty set
$\sim$	Relates to
$>$	Greater than
$<$	Less than
$\geq$	Greater than or equal to
$\leq$	Less than or equal to
$ n $	The absolute value of n
$\square$	It has been proven

Part I

Essentials of Multivariable Calculus

Chapter 1

Vectors, Vector Valued Functions, and Functions of Several Variables

1.1 The Vector Space $\mathbb{R}^n$

A VECTOR SPACE V on $\mathbb{F}$ is the set V of objects named *vectors* and a field¹ $\mathbb{F}$ of objects name *scalars*, with two operations *vector addition* and *scalar multiplication* defined:

1. Vector Addition: $+: V \times V \rightarrow V$
2. Scalar Multiplication: $\cdot: \mathbb{F} \times V \rightarrow V$

That satisfies the axioms of a vector space. Namely that vector addition and scalar multiplication are both commutative, associative, and distributive, and that there exists additive and multiplicative identities and inverses.

THE VECTOR SPACE $\mathbb{R}^n$ is the set $V = \{(x_0, x_1, \dots, x_n) : x_i \in \mathbb{R}\}$ over the real numbers. Elements of V are said to be n -tuples of real numbers. Vector addition is defined to be the coordinate-wise addition of elements of V :

$$(x_0, x_1, \dots, x_n) + (x'_0, x'_1, \dots, x'_n) = (x_0 + x'_0, x_1 + x'_1, \dots, x_n + x'_n)$$

with the usual arithmetic inherited from the real numbers. Scalar multiplication is defined as the distributive multiplication of each coordinate by some $\alpha \in \mathbb{R}$:

$$\alpha \cdot (x_0, x_1, \dots, x_n) = (\alpha \cdot x_0, \alpha \cdot x_1, \dots, \alpha \cdot x_n)$$

In two and three dimensions, n -tuples of numbers can be thought of as a quantity with *magnitude and direction*, and arrow from the origin to (x, y, z) , however to not lose generality in higher dimensions it is best just to think of them as ordered lists of real numbers, with arithmetic defined above.

¹ refer: Algebra

THE DOT PRODUCT of two vectors is defined:

$$\mathbf{a} \cdot \mathbf{b} = \sum_{n=1}^n a_i b_i$$

for two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$. It is also known as the scalar or inner product. An alternative formulation of the dot product is:

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$$

where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$, and $\|\mathbf{a}\|$ is the magnitude, or length, of $\mathbf{a}$. Thus, two orthogonal vectors will have a dot product of 0 (since $\cos(\pi/2) = 0$) and two parallel vectors will have a dot product equal to the product of their magnitudes. Notice

$$\mathbf{a} \cdot \mathbf{a} = \|\mathbf{a}\|^2 \implies \|\mathbf{a}\| = \sqrt{\mathbf{a} \cdot \mathbf{a}}$$

The dot product can be thought of geometrically as a measure of how much of $\mathbf{a}$ goes in the direction of $\mathbf{b}$. A handy application of this fact is determining the x, y , and z components of a vector $\mathbf{a}$, by taking its dot product with the standard basis vectors² $\mathbf{i}, \mathbf{j}$, and $\mathbf{k}$, respectively.

THE CROSS PRODUCT of two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$ is defined:

$$\mathbf{a} \times \mathbf{b} = (\|\mathbf{a}\| \|\mathbf{b}\| \sin \theta) \cdot \mathbf{n}$$

where θ is the angle between them and $\mathbf{n}$ is a unit vector perpendicular to the plane containing $\mathbf{a}$ and $\mathbf{b}$, whose direction is given by the right hand rule (fig. 1.1). A vector perpendicular to a plane is said to be *normal* to that plane.

An easy way to calculate the cross product is to build the 3×3 matrix with first row $[\mathbf{i} \ \mathbf{j} \ \mathbf{k}]$, second row $[\mathbf{a}]$, and third row $[\mathbf{b}]$, and take it's determinant.

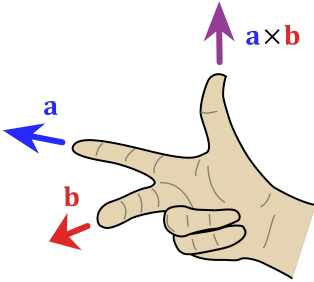


Figure 1.1: The right hand rule is the convention used for the positive direction $\mathbf{n}$ in $\mathbf{a} \times \mathbf{b}$.

² $\mathbf{i} = (1, 0, 0)$, $\mathbf{j} = (0, 1, 0)$, $\mathbf{k} = (0, 0, 1)$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$\mathbf{a} \times \mathbf{b} = (a_2 b_3 - a_3 b_2)\mathbf{i} + (a_3 b_1 - a_1 b_3)\mathbf{j} + (a_1 b_2 - a_2 b_1)\mathbf{k}$$

Once you know a normal vector to a plane (usually by taking the cross product of two vectors in that plane) and one point in the plane, it is easy to determine an equation for that plane, since the dot product of *any* vector in the plane with the normal vector will be 0.

$$\mathbf{n} \cdot (x, y, z) = n_x x + n_y y + n_z z + d = 0$$

Simply plug in the (x, y, z) that you know to determine the correct value of d .

1.2 Vector Valued Functions

A FUNCTION f is said to *vector valued* if it is a mapping $\mathbb{R} \rightarrow \mathbb{R}^n$. In other words, vector valued function take a real number and assign a vector.

$$\mathbf{r}(t) = (x_1(t), x_2(t), \dots, x_n(t))$$

The functions $x_1(t), \dots, x_n(t)$ are called the *coordinate functions* of $\mathbf{r}$.

A common example of a vector valued function is $\mathbf{r}(t) = (t, f(t))$ where $f : \mathbb{R} \rightarrow \mathbb{R}$ and $t \in [a, b] : a, b \in \mathbb{R}$. This function describes a *parametrisation* of f and points from the origin to every point of f as t varies in its domain, as if the tip were moving along f . It is helpful to imagine t as representing time. This will be covered in more detail in Chapter 7.

1.3 Functions of Several Variables

A FUNCTION f is a *multivariable* function if it is a mapping from $\mathbb{R}^n \rightarrow \mathbb{R}$. These functions take a vector and assign to it a real number. The coordinates of the input vector can be thought of as several independent variables.

$$f(x_1, x_2, \dots, x_n) = n$$

where n is some combination of the variables $x_1, \dots, x_n$. Just like we can plot a function of one variable $f : D \rightarrow \mathbb{R}$ by defining the line $l = \{(x, f(x)) : x \in D\}$ as the set of all points that satisfy $(x, f(x))$, we can define an $n + 1$ dimensional surface of a function $f : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R} \subseteq \mathbb{R}$ of n variables by

$$S = \{(x_1, x_2, \dots, x_n, f(x_1, x_2, \dots, x_n)) : x_1, \dots, x_n \in D\}$$

Allowing an $n + 1$ dimensional alien to understand its plot. The only applications where this is useful to humans is to plot $z = f(x, y)$ or every point (x, y, z) that satisfies $f(x, y, z) = k$ for some $k \in \mathbb{R}$ in three dimensions.

FIELDS are mappings $\mathbb{R}^n \rightarrow \mathbb{R}^m$, usually $\mathbb{R}^3 \rightarrow \mathbb{R}^m$. They can be thought of as assigning to every point in space a vector in $\mathbb{R}^m$. A common example is a force or electromagnetic field. A scalar field is a special multivariable function of three variables that assigns to every point in space a number, like temperature or pressure. These are covered in Chapter 8.

Chapter 2

Limits and Continuity

2.1 Functions of One Variable

A FUNCTION f approaches a limit l near a if for every $\epsilon > 0$, there exists some $\delta > 0$, such that if $0 < |x - a| < \delta$, then $|f(x) - l| < \epsilon$.

This is a precise way of writing that if a function approaches a value l near a , then no matter how small of an interval we draw around l , we will always be able to draw an interval around a , where $f(x)$ is within that interval of l . It will become clear in the examples why this definition matches the intuitive understanding of what a limit should be.

CONSIDER $f(x) = x^2$ (fig. 2.1). Obviously it approaches 0 near 0 (and a^2 near a), but how would we show that? We have to decide how to ensure that $|x^2 - a^2| < \epsilon$ whenever $|x - a| < \delta$. We want

$$|x^2 - a^2| = |x - a| \cdot |x + a| < \epsilon$$

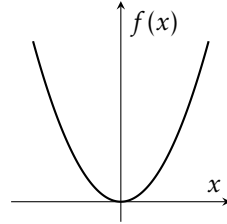


Figure 2.1: $f(x) = x^2$.

The factor $|x + a|$ is obviously the problem, but as long as we know an upper bound M for it, we can just ensure that $|x - a| < 1/M$. If we take $|x + a| < 1$, then

$$|x + a| = |(x - a) + 2a| \leq |x - a| + |2a| < 1 + 2|a|$$

Now, we just need

$$|x - a| < \frac{\epsilon}{1 + 2|a|}$$

for $|x^2 - a^2| < \epsilon$. So, let $\epsilon > 0$, choose $\delta = \min\left(1, \frac{\epsilon}{1 + 2|a|}\right)$. Then $\delta \leq 1$, so $|x - a| < 1$ and $|x + a| < 1 + 2|a|$. Consider $|x^2 - a^2| = |x - a| \cdot |x + a|$. Substituting our bounds, $|x^2 - a^2| = \delta \cdot (1 + 2|a|) \leq \frac{\epsilon}{1 + 2|a|} \cdot (1 + 2|a|) = \epsilon$.

So for every ϵ , there is an interval around a , equal to $a \pm \min\left(1, \frac{\epsilon}{1 + 2|a|}\right)$ that will ensure $|f(x) - l| < \epsilon$.

CONSIDER $f(x) = \sin(1/x)$. (fig. 2.2)

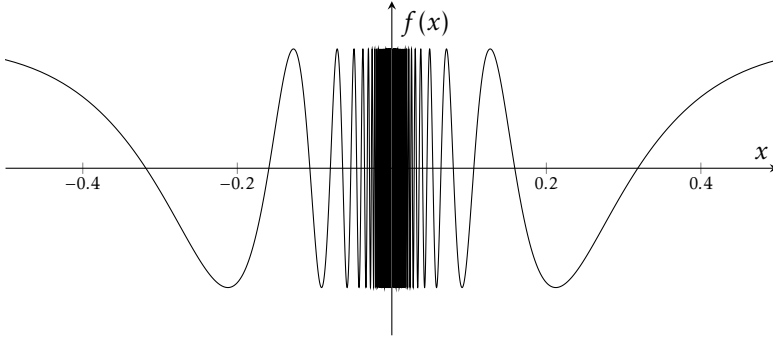


Figure 2.2: $f(x) = \sin(1/x)$.

This function oscillates wildly as $x \rightarrow 0$ and so does not approach any value at $x = 0$. The intuition for this fact is that no matter how small we make an interval containing 0, we will always be able to choose a large enough natural number n , such that there is a point $a = \frac{2}{\pi(1 + 4n)}$ where $\sin a = 1$, and a point $b = \frac{2}{\pi(3 + 4n)}$ where $\sin b = -1$.

Assume for contradiction that $\lim_{x \rightarrow 0} \sin 1/x = l$. Let $\epsilon = 1/2$ and $\delta > 0$. Now choose a sufficiently large natural number n , such that $a = \frac{2}{\pi(1 + 4n)} < \delta$.

Then, there will also be the point $b = \frac{2}{\pi(3 + 4n)} < a < \delta$. Then,

$$\begin{aligned} |f(a) - l| &= |1 - l| < \epsilon = \frac{1}{2} \\ |f(b) - l| &= |-1 - l| < \epsilon = \frac{1}{2} \end{aligned}$$

By the triangle inequality

$$|1 + 1| = |(1 - l) + (1 + l)| \leq |1 - l| + |1 + l| < \frac{1}{2} + \frac{1}{2} = 1 \implies 2 < 1$$

Which is a contradiction. So our original assumption that the limit exists and equals l must be wrong. $\square$

A SLIGHTLY MORE COMPLICATED EXAMPLE.

$$f(x) = \begin{cases} 0 & x \text{ irrational, } 0 < x < 1 \\ \frac{1}{q} & x = \frac{p}{q} \text{ in lowest terms, } 0 < x < 1 \end{cases}$$

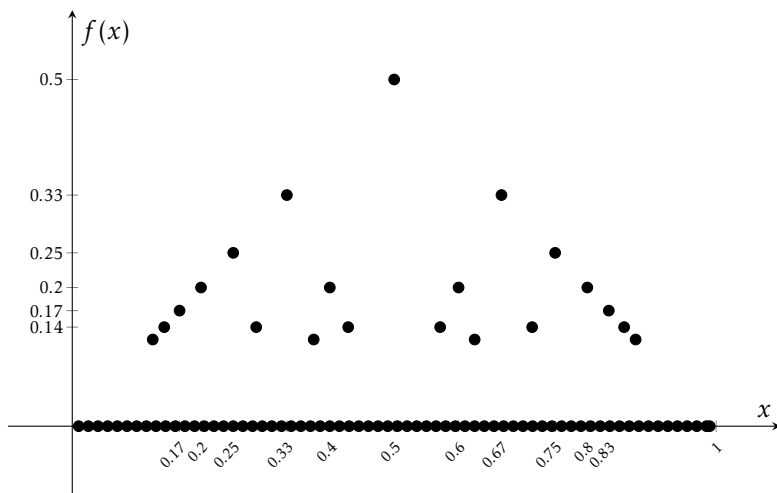


Figure 2.3: $f(x)$.

Perhaps counterintuitively at the moment, this function approaches 0 at a . Recall the definition of a limit, the function f *approaches* l at a if for every $\epsilon > 0$, $\exists \delta > 0$, with $0 < |x - a| < \delta$, and $|f(x) - l| < \epsilon$. We don't actually care what $f(a)$ is, since the difference between x and a will always be greater than 0, the value at $f(a)$ is irrelevant.

Let $\epsilon > 0$ be given. Choose a sufficiently large natural number n such that

$1/n < \epsilon$. The only numbers x where $|f(x) - 0| < \epsilon$ could be false are

$$\frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{3}{4}, \frac{1}{5}, \frac{2}{5}, \dots, \frac{1}{n}, \dots, \frac{n-1}{n}$$

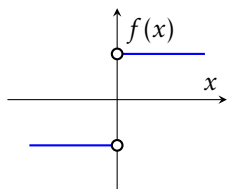
the set of which we can call S .

However many of these numbers there are, there are only finitely many, and so one of them is closest to a . That is, $|p/q - a|$ is smallest for one specific $p/q \in S$. If a is in S , then just consider all the elements of $S \neq a$. If we chose δ to be this smallest distance then for all x satisfying $0 < |x - a| < \delta$, x is guaranteed not to be in S , and therefore $|f(x) - 0| < \epsilon$, which completes our proof. $\square$

There is no need to give δ in terms of ϵ , since there is nowhere in the definition $\dots \forall \epsilon, \exists \delta \dots$ that says each ϵ needs a *unique* δ .

2.1.1 Divergence to, and Limits at, Infinity

THE DEFINITION OF A LIMIT can be modified to handle divergence to infinity. A function l approaches ∞ near a if for every $M > 0$, there exists a δ such that, whenever $|x - a| < \delta$, $f(x) > M$. No matter how large M is, there will always be an interval we can draw around a , such that $f(x) > M$.



LIMITS AT INFINITY have a similar modified definition. A function f approaches a limit l near ∞ if for every $\epsilon > 0$, there exists an $M > 0$ such that if $x > M$, $|f(x) - l| < \epsilon$. For every interval around the limit we draw, there is always a point on the number line that ensures we are within that interval.

Figure 2.4: The derivative of the absolute value function.

2.1.2 One Sided Limits

ONE SIDED LIMITS are exactly the same as the full limit, except that x is only considered to be approaching a from the left or the right. The full limit only exists if both one sided limits equal, but it is possible for either one sided limit to exist independently.

For example, define (fig. 2.4)

$$f(x) := \frac{d|x|}{dx} = \begin{cases} -1 & x < 0 \\ 1 & x > 0 \end{cases}$$

If we examine what f does as $x \rightarrow 0$ from the left, its quite obvious that it approaches -1, and 1 from the right. We say that

$$\begin{aligned} \lim_{x \rightarrow 0^-} f(x) &= -1 \\ \lim_{x \rightarrow 0^+} f(x) &= 1 \end{aligned}$$

but the full limit:

$$\lim_{x \rightarrow 0} f(x)$$

is not defined.

2.2 Vector Valued Functions

WE GET THESE FOR FREE. Just define

$$\lim_{t \rightarrow a} \mathbf{r}(t) := \left(\lim_{t \rightarrow a} x_1(t), \lim_{t \rightarrow a} x_2(t), \dots, \lim_{t \rightarrow a} x_n(t) \right)$$

If the limits of the coordinate functions exist, then they can be evaluated using the same techniques as single variable functions

2.3 Techniques for Evaluating Limits

THE LIMIT LAWS allow for breaking up complicated limits in several trivial to solve limits.

1. $\lim_{x \rightarrow a} (f + g)(x) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$
2. $\lim_{x \rightarrow a} f(x) \cdot g(x) = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$
3. $\lim_{x \rightarrow a} c \cdot f(x) = c \cdot \lim_{x \rightarrow a} f(x)$
4. $\lim_{x \rightarrow a} \left(\frac{f(x)}{g(x)} \right) = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$

5. If $\lim_{x \rightarrow a} g(x) = l$ and f is continuous at l , then $\lim_{x \rightarrow a} f(g(x)) = f(l)$

THE SQUEEZE THEOREM states that if $g(x) \leq f(x) \leq h(x)$ and $\lim_{x \rightarrow a} g(x) = l = \lim_{x \rightarrow a} h(x)$, then $\lim_{x \rightarrow a} f(x) = l$.

L'HÔPITAL'S RULE states that if f and g are differentiable, and $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$ or $\pm\infty$, then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$.

Proofs of these assertions are a bit beyond the scope of these notes.

2.4 Limits of Multivariable Functions

FOR LIMITS of multivariable functions, one must consider every possible path to a . Just like with functions of one variable, where we require the left and right sided limits to equal, for multivariable functions, we require all paths to be equal. For example:

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \frac{xy}{x^2 + y^2}$$

Consider the path $x = y$:

$$\lim_{(x,y) \rightarrow (x,x)} f(x,y) = \lim_{(x,y) \rightarrow (x,x)} \frac{x^2}{x^2 + x^2} = \frac{1}{2}$$

Now, consider the path $x = 0$:

$$\lim_{(x,y) \rightarrow (0,y)} f(x,y) = \lim_{(x,y) \rightarrow (0,y)} \frac{0}{y^2} = 0 \neq \frac{1}{2}$$

Thus, the limit does not exist.

The technique for solving limits of multivariable functions is to investigate a few paths to observe their behaviour. If the limit exists then a symbolic proof must be given to cover all possible paths. Otherwise, one example will suffice. Evaluating the limit uses similar techniques to single variable limits.

Consider

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + x + xy + y}{x + y}$$

Naively evaluating $(0,0)$ produces an indeterminate $\frac{0}{0}$, but L'Hôpital's rule is not defined for multivariable functions. However if we factorise the numerator:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + x + xy + y}{x + y} = \lim_{(x,y) \rightarrow (0,0)} \frac{(x+y)(x+1)}{x+y} = \lim_{(x,y) \rightarrow (0,0)} x+1 = 1$$

We did not make any assumptions about the path taken by x and y to $(0,0)$, and so this symbolic proof remains true for every possible path.

2.5 Continuity

A FUNCTION IS CONTINUOUS if $\lim_{x \rightarrow a} f(x) = f(a)$. Common continuous functions include

1. All polynomials $p_0 + p_1x + p_2x^2 + \dots + p_nx^n$.
2. All rational functions $R(x) = \frac{P(x)}{Q(x)}$ except where $Q(x) = 0$.
3. $\sin x$ and $\cos x$.
4. e^x and $\ln x$ (on its domain $x \in (0, \infty)$).
5. x^α for $\alpha \in \mathbb{R}$ and $x > 0$.
6. Linear combinations and products of continuous functions are continuous.
7. Ratios of continuous functions are continuous, provided the denominator is not 0.
8. $f(g(x))$ is continuous when $g(x)$ is continuous, provided the codomain of g is the domain of f .

Chapter 3

Derivatives

FOR A FUNCTION f , define its *derivative* at a :

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Chapter 4

Applications of the Derivative

Chapter 5

The Riemann Integral

Chapter 6

Double and Triple Integrals

Chapter 7

Path and Surface Integrals

Chapter 8

Vector Fields